Name & Surname **Polimi ID**

Exercise 1

A. Describe the reduce parallel pattern and at least one fusion optimization that can be applied to it. (1)

B. Please provide a practical example of instruction dependency that prevents parallelization and an example that does not prevent parallelization. (1)

C. Given the following C-based code:

```
int sum = 0;

for (i = 2; i < 100; i++) {
   for (j = 1; j < 100; j++) {
      x[i][j] = g(x[i-2][j], y[i][j-1]);
   }
}

for (i = 0; i < 100; i++) {
   for (j = 1; j < 100; j++) {
      z[i][j] = z[i][j] + x[i][j] * w[i][j];
      sum += z[i][j];
   }
}
```

where matrices x, y, z, and w store integers and already contain valid data before the execution of the code.
Discuss how and under which conditions the previous code can be parallelized. In particular, identify possible loop-carried dependencies, specify which loops can be safely parallelized, and state which OpenMP pragmas you would use. Discuss how the variable sum must be handled. (2)

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Exercise 2

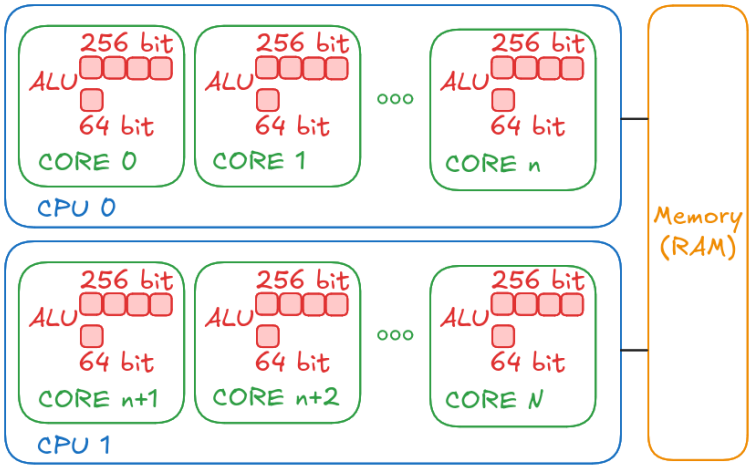
A. Please provide a short pseudocode for the parallel scatter pattern. Discuss its assumptions and PRAM complexity. In which other parallel patterns discussed during the lecture has the scatter been used? (2)

B. Please describe how the Pack parallel pattern works and its performance considering a PRAM model. How is it implemented (pseudo code required)? (2)

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Exercise 3

A. Based on the architecture illustrated in the figure below, describe how you would exploit the different levels of parallelism to accelerate the execution of an application. (1)



B. When is it useful to use the OpenMP `sections` construct instead of tasking? Provide an explanation to support your answer. (1)

C. Based on the architecture illustrated in point A, how would you parallelize the following code snippet using OpenMP pragmas? (2)

```
float a[N], b[N]; // initialization of a and b is omitted
float c = 0.0f;

for (int i = 0; i < N; i++) {

    c += (a[i] + b[i]) * (a[i] + b[i]) + a[i] * 3.0f;

}
```

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Exercise 4

A. Assuming that `echo` is a serial executable that prints its command-line arguments to standard output, what would be the output of the following command? `mpirun –np 4 echo hello world?` (1) **This question can be skipped if you got 1 or 3 points in the second challenge.**

B. If I were to look inside MPI, what would I expect to read? Can you give some examples? (1)

C. Describe how you could use OpenMP to parallelize the snippet of pseudocode written below that uses non-blocking semantics to overlap communication with computation (2) **This question can be skipped if you got 2 or 3 points in the second challenge.**

```
MPI_Irecv_c(/* parameter recv 1 */, &request);
int request_is_complete = 0;
while(request_is_complete == 0) {
 do_work(); // this function perform the computation
  MPI_Test(&request, &request_is_complete, /* the status of recv */)
}
```